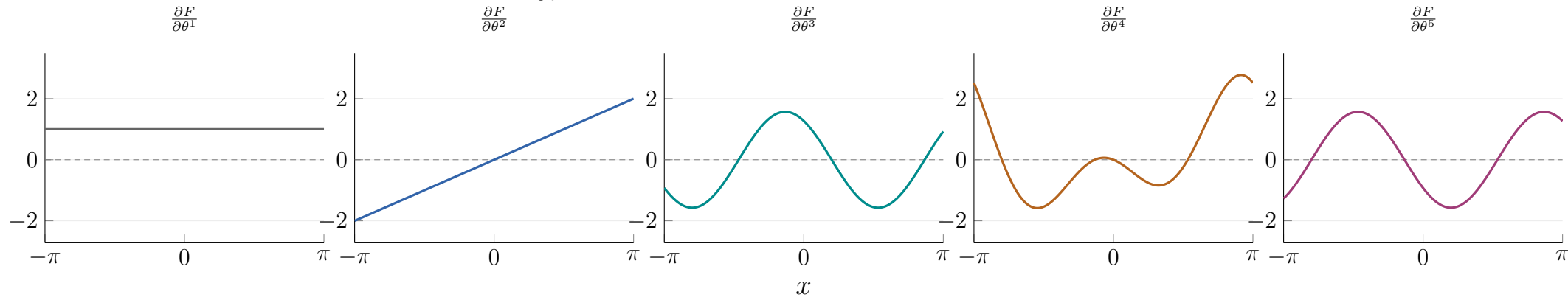
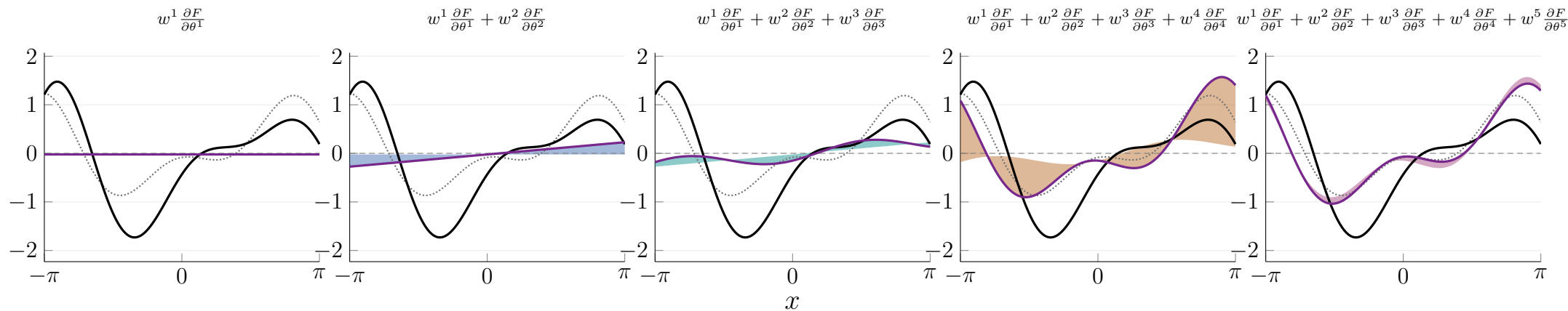


Partial derivatives $\frac{\partial F}{\partial \theta^i}$ and the update direction built one parameter at a time

a.



b.



$\cdots \Delta f$ (achieved update) $\text{shaded band: contribution of } k\text{-th partial derivative}$
 $\text{--- } \nabla_f P = c$ (functional gradient) $\text{--- } \sum_{i=1}^k w^i \frac{\partial F}{\partial \theta^i}(x)$ (First-order approximation of update)